Phase I Crossover Bioavailability Study of Tomovicen, a Novel Tamoxifen-like Prodrug

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Abstract

Tomovicen is an investigational tamoxifen-like prodrug developed to improve on the pharmacokinetic (PK) and pharmacodynamic (PD) profile of tamoxifen. This study characterised its bioavailability and primary PK parameters. Twelve healthy adults (8 female and 4 male; aged 50–64 years) received 500 mg as a 0.5 h intravenous (IV) infusion and 500 mg orally in a two-period crossover. PK parameters were derived by non-compartmental analysis (NCA) in PKNCA and PKanalix, with a covariate screening and a proposed compartmental model also assessed. Clearance was low (CL 0.96 ± 0.19 L/h) and distribution limited (Vz 8.7 ± 0.46 L), yielding a half-life of 6.5 ± 0.96 h. Mean absolute bioavailability was 69% (range 45%–91%). Body weight and the provisionally grouped CYP2D6 categories emerged as candidate covariates for further investigation, and a one-compartment model was selected for both routes. Tomovicen is moderately bioavailable with a short half-life, predicting negligible once-daily accumulation. Suitability for once-daily dosing requires metabolite measurement and a pharmacodynamic target. Tomovicen is designed to address these limitations while retaining the anti-oestrogenic mechanism of action. This Phase I two-sequence, two-period crossover study in healthy volunteers determined the absolute oral bioavailability an primary PK parameters of the parent compound, providing the foundation for dose selection in Phase II.

1 Introduction

Tamoxifen, a non-steroidal selective oestrogen receptor modulator (SERM), is a widely used endocrine therapy for ER-positive breast cancer (adjuvant, advanced and male disease; 20 mg/day). It is an ER antagonist in breast tissue and a partial agonist in the endometrium, bone and liver, which explains both its benefits (bone mass, lower LDL cholesterol) and its adverse effects (hot flushes, thromboembolism, endometrial cancer).("Tamoxifen," n.d.)

After oral dosing, tamoxifen is rapidly absorbed (t_{max} 4–7 h; C_{max} about 40 ng/mL), highly protein-bound (>99%) and widely distributed (about 50–60 L/kg). It is a prodrug: CYP3A4/5 converts it to *N*-desmethyltamoxifen, then CYP2D6 to endoxifen, whose ER affinity is 30–100 times higher than that of the parent. Elimination is slow (half-life 5–7 days for tamoxifen, about 14 days for *N*-desmethyltamoxifen), so steady state takes about three months ("Tamoxifen: Uses, Interactions, Mechanism of Action," n.d.).

This pharmacology raises several challenges, pharmacokinetically, it is a prodrug that depends on CYP2D6 for activation, so exposure to endoxifen varies widely between patients (poor metabolisers and strong CYP2D6 inhibitors lower it by 65-75%). Its very long half-lives (5-14 days) and enterohepatic recirculation delay steady state and complicate modelling, and its oral-only use, with high protein binding, prevents estimating absolute bioavailability, clearance and volume of distribution. Pharmacodynamically, its mixed agonist/antagonist activity causes both efficacy and adverse effects, its response depends on active metabolites rather than on the dose given, and it is limited by resistance, poor adherence and a poorly defined therapeutic window (endoxifen threshold of about 5.97 ng/mL, still debated).

2 Methods

2.1 Study design

Twelve participants received 500 mg Tomovicen as a 0.5 h IV infusion and 500 mg orally in a randomised sequence, separated by a washout of undocumented length. Plasma samples were collected at 17–18 nominal time points from 0.5 to 96 h after each dose; only the parent compound was measured.

2.2 Data management

The raw dataset (one Excel sheet per participant) was audited and curated without modifying the source file; all anomalies and their resolutions are documented in the companion interactive dashboard report. Data management followed the FAIR principles: scripted data transformations, a variable dictionary and version-controlled code to support findability, accessibility, interoperability and reusability. eGFR was calculated with the CKD-EPI 2021 equation.

2.3 Pharmacokinetic analysis

NCA was performed with PKNCA (Denney, Duvvuri, and Buckeridge, n.d.) in R (version 4.5.1). AUC was computed by the linear-up/log-down trapezoidal rule; the terminal rate constant (λ_z) was estimated by log-linear regression on ≥ 3 post-peak points ($R^2 \geq 0.90$; extrapolated area $\leq 20\%$). The absorption rate constant (k_a) was derived by the method of residuals and the absolute bioavailability as $F = (AUC_{oral}/D_{oral})/(AUC_{IV}/D_{IV})$ within each participant. The once-daily accumulation ratio was predicted as $R = 1/(1 - e^{-\lambda_z \tau})$ ($\tau = 24$ h) and the steady-state trough-to-peak ratio as $e^{-\lambda_z \tau}$, assuming linear kinetics. Period and carry-over effects were tested on log $AUC_{0-\infty}$ with Welch t-tests between the two sequences, using within-participant period differences and participant totals, respectively. Compartmental analysis was performed in MonolixSuite PKanalix 2024R1. Covariate screening used individual PKanalix NCA parameters (adjusted $R^2 \geq 0.98$), with Spearman rank correlation (ρ) for continuous covariates and the Wilcoxon rank-sum test for categorical covariates, with CYP2D6 categories 1–2 (normal function) pooled against 3 and 5 (no function); a value was considered statistically significant when $p < 0.05$.

3 Results

3.1 Baseline characteristics

Table 1. Baseline characteristics of the study cohort (n = 12). CYP2D6 codes 1–2: normal-function; 3 and 5: no-function alleles.

(a) Continuous

Characteristic	Mean ± SD	Range
Age (years)	56.8 ± 4.6	50.3–64.4
Weight (kg)	77.6 ± 9.2	64.4–92.9
eGFR (mL/min/1.73 m²)	87.1 ± 19.3	40.5–107.5
Albumin (g/L)	45.0 ± 2.7	40.6–50.0
CRP (mg/L)	2.9 ± 1.7	0.3–6.1

(b) Categorical

Characteristic	n (%)
Female / male	8 (67) / 4 (33)
CYP2D6 code 1 / 2	3 (25) / 3 (25)
CYP2D6 code 3 / 5	2 (17) / 4 (33)
Asian / Black	3 (25) / 3 (25)
Other / White	3 (25) / 3 (25)

The 12 participants were middle-aged adults with mean body weight 77.6 kg and mean eGFR 87.1 mL/min/1.73 m²; one participant had markedly reduced renal function (eGFR 40.5 mL/min/1.73 m²). Eight participants were female and four male, balanced across the two treatment sequences. All continuous covariates were consistent with normality (Shapiro–Wilk $p > 0.09$) except creatinine ($p = 0.04$), which was influenced by the same participant. Each CYP2D6 subtype and ethnic group contained only two to four participants, limiting subgroup comparisons.(Wilkinson 2016)

3.2 Non-compartmental analysis

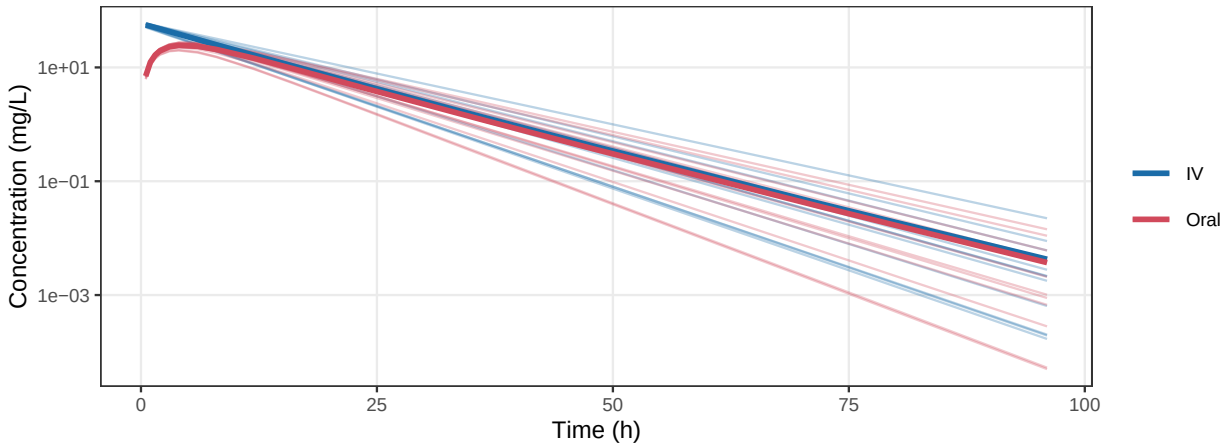


Figure 1: Plasma concentration–time profiles after 500 mg IV infusion and 500 mg oral dose (semi-logarithmic scale). Thin lines: individual participants; thick lines: arithmetic mean.

All 24 profiles met the acceptance criteria ($R^2 \geq 0.9999$; extrapolated area $< 0.04\%$). After IV infusion, concentrations declined as a single log-linear phase with no visible distribution phase, indicating one-compartment disposition. After oral dosing, concentrations peaked at approximately 4 h and the similar terminal slopes after oral and IV administration suggest that the terminal decline after oral dosing is governed by drug elimination, not by slow absorption

Table 1: Pharmacokinetic parameters of Tomovicen after 500 mg IV infusion and 500 mg oral dose (n = 12). Values are mean ± SD unless marked; geometric CV = $100\sqrt{e^{s^2_{\log}} - 1}$. [a]Median [IQR]. [b]Implausibly uniform estimate from the method of residuals (see Limitations). F: within-participant dose-normalised AUC ratio.

Parameter	Mean ± SD	Geom. mean	Geom. CV (%)	Range
IV infusion				
C_{max} (mg/L)	55.9 ± 3	55.9	5	50.4–59
t_{max} (h)[a]	0.5 [0.5–0.5]	–	–	0.5–0.5
$AUC_{0-\infty}$ (mg·h/L)	540 ± 103	531	19	400–731
CL (L/h)	0.958 ± 0.19	0.941	19	0.684–1.25
V_z (L)	8.72 ± 0.46	8.71	5	8.28–9.61
V_{ss} (L)	8.73 ± 0.46	8.72	5	8.28–9.61
$t_{1/2}$ (h)	6.48 ± 0.96	6.41	15	5.24–8.4
MRT (h)	9.35 ± 1.4	9.26	15	7.57–12.1
Oral				
C_{max} (mg/L)	24.3 ± 2.6	24.2	11	19.8–27.2
t_{max} (h)[a]	4 [4–4]	–	–	4–4
$AUC_{0-\infty}$ (mg·h/L)	370 ± 84	361	24	242–490
CL/F (L/h)	1.42 ± 0.35	1.39	24	1.02–2.06
V_z/F (L)	12.7 ± 0.87	12.7	7	11.7–14.3
k_a (h^{-1})[b]	0.43 ± 0.0033	0.43	1	0.425–0.436
$t_{1/2}$ (h)	6.43 ± 1.1	6.34	18	4.81–8.12
MRT (h)	11.8 ± 1.6	11.7	14	9.42–14.2
Bioavailability				
F	0.694 ± 0.15	0.679	22	0.454–0.914

The compound shows low clearance (CL 0.96 L/h, ≈ 1% of hepatic blood flow), making exposure sensitive to enzyme activity rather than blood flow. Volume of distribution was limited ($V_z \approx V_{ss} \approx 8.7$ L, well below total body water),

unlike tamoxifen. The half-life was the same by both routes (6.5 vs 6.4 h; ratio 1.00, 95% CI 0.90–1.10) and no statistically significant period or carry-over effect was detected ($p > 0.25$). Between-subject variability was moderate for clearance and exposure (CV 19–25%) but low for volume ($\approx 5\%$). Assuming linear kinetics, once-daily dosing would give a predicted accumulation ratio of 1.08, with steady-state trough concentrations at approximately 7% of peak.

3.3 Absolute bioavailability

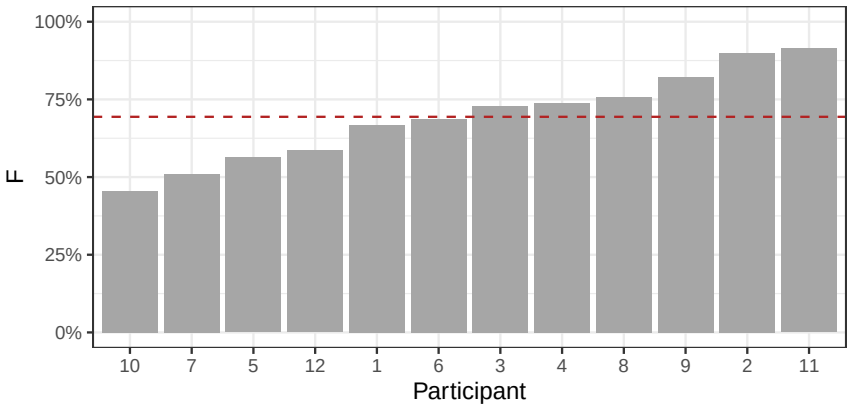


Figure 2: Absolute oral bioavailability (F) per participant (ranked). Dashed line: arithmetic mean.

Mean absolute oral bioavailability was 69% (SD 15%; range 45%–91%), and every participant had lower exposure after the oral than after the IV dose (Figure 2). About one third of the oral dose therefore does not reach the systemic circulation, which may reflect incomplete absorption or gut-wall metabolism; given the low hepatic extraction suggested by the clearance, hepatic first-pass loss is expected to be small. Whether part of this loss corresponds to conversion to the active metabolite cannot be assessed, as only the parent compound was measured. F also varied noticeably between individuals. This variability may reflect differences in gastrointestinal absorption, such as gastric emptying, intestinal transit and gut-wall metabolism, and possibly in CYP2D6-mediated conversion of the prodrug.

3.4 Covariate pilot analysis

Covariate relationships were explored separately for IV infusion and oral administration using individual NCA parameters from PKanalix. Visual inspection of covariate–parameter plots was complemented by Spearman rank correlations for continuous covariates and Wilcoxon rank-sum tests for categorical covariates, calculated in R. CYP2D6 categories {1,2} and {3,5} were pooled, conditional on the dataset codes representing the corresponding normal-function and no-function star alleles. Findings were interpreted according to association strength, consistency across related parameters and biological plausibility, rather than nominal p-values alone.

After IV dosing, body weight was the most consistent covariate: heavier participants had lower C_{max} (Spearman $\rho = -0.52$, $p = 0.089$) and larger V_z ($\rho = +0.49$, $p = 0.11$). The eGFR and creatinine associations may partly reflect body-size differences rather than a direct renal effect. A plausible albumin–k_e association ($\rho = -0.40$, $p = 0.20$) may reflect protein-binding effects on clearance. After oral dosing, higher CRP was associated with lower $AUC_{0-\infty}$ ($\rho = -0.60$, $p = 0.043$) and C_{max} ($\rho = -0.65$, $p = 0.026$), potentially reflecting inflammation-related changes in oral bioavailability. Across both routes, the pooled CYP2D6 {3,5} group tended to have higher exposure and a slower terminal decline than the {1,2} group; the pooled comparison reached nominal significance for oral $AUC_{0-\infty}$ and k_e ($p = 0.041$), consistent with CYP2D6 contribution to clearance. These findings remain exploratory given the small sample size and the number of associations examined.

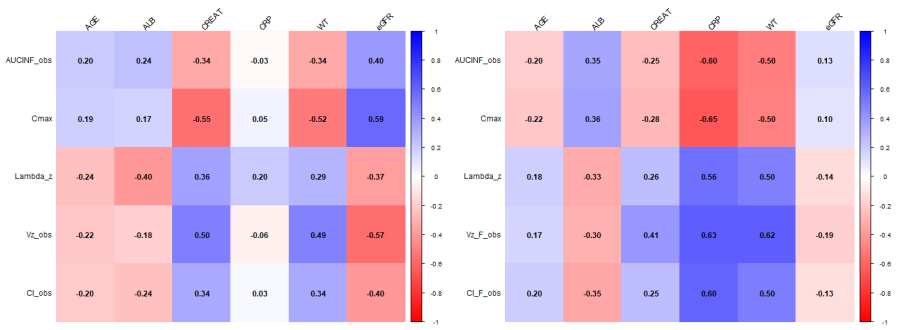


Figure 3: Spearman rank correlation (ρ) between IV PK parameters (left), Oral PK parameters (right) and continuous covariates.

3.5 Proposed compartmental model

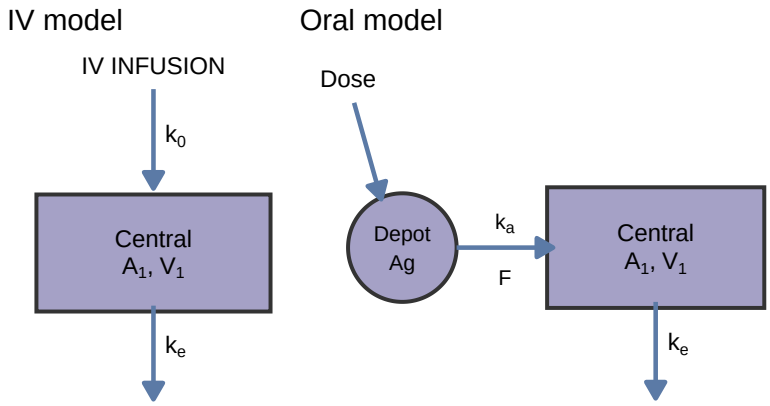


Figure 4: One-compartment model diagrams. Left(IV): zero-order infusion (k0) with first-order elimination (ke). Right(Oral): first-order absorption (ka) with bioavailability fraction F and first-order elimination (ke).

$$\frac{dA_1}{dt} = k_0 - k_e A_1(t), \qquad \frac{dA_{\mathrm{IV}}}{dt} = k_0 \tag{1}$$

$$\frac{dA_g}{dt} = -k_a A_g(t), \qquad \frac{dA_1}{dt} = F\, k_a A_g(t) - k_e A_1(t) \tag{2}$$

A_{IV} is the cumulative amount infused; k_0 is 1000 mg/h during the 0.5 h infusion and zero thereafter. For the oral model, $A_g(0) = \mathrm{Dose}$ and $A_1(0) = 0$.

Individual concentration–time profiles after IV infusion showed a single log-linear decline with $V_z \approx V_{ss}$, supporting a one-compartment structural model with linear elimination. Oral profiles declined in parallel to the IV curve, confirming first-order absorption without a lag time. A one-compartment model with zero-order infusion (IV) and first-order absorption with bioavailability F (oral) was fitted in PKanalix; individual fits lay close to the line of identity for both routes, supporting a one-compartment model. Initial parameter estimates taken from Table 2 confirmed good convergence.

4 Discussion

Only the parent compound was measured in this study. Tomovicen is designed as a prodrug, and the active moiety responsible for anti-oestrogenic effect has not been identified or quantified. Without metabolite data it is not possible to confirm CYP2D6-mediated activation, to link exposure of the parent to pharmacodynamic effect, or to judge whether the trough concentration at 24 h (approximately 7% of peak for once-daily dosing) remains above an efficacious level. Second, the high degree of protein binding characteristic of tamoxifen-like structures may also apply to Tomovicen; if protein binding is substantial, free (pharmacologically active) drug concentrations will differ from total plasma concentrations, and inter-individual variability in albumin could produce clinically relevant differences in free exposure. Third, the washout duration between periods is undocumented, so carry-over cannot be formally excluded although both the period and carry-over tests were non-significant ($p > 0.25$). Finally, the dose and dosing interval cannot be set on PK grounds alone: both require a pharmacodynamic target for the active species, which needs dose–response data and quantification of conversion to the active metabolite(s).

Limitations

The sample of 12 participants provides low statistical power; many of the covariate associations identified are likely to reflect chance rather than true effects. CYP2D6 subtype, sex and ethnicity are partially confounded, so these covariates cannot be assessed independently. The CYP2D6 grouping assumed that the numeric codes correspond to specific star alleles under the CPIC classification, which was not verified. The very low between-participant variability in estimated ka warrants verification of the residual-method calculations and their sensitivity to sampling times. Concentration data for three participants relied on scale corrections not yet confirmed by the data manager. The covariate analysis used individual parameters from PKanalix, which may differ slightly from the PKNCA values in Table 2.

5 Conclusion

Tomovicen displays low clearance (0.96 L/h), limited volume of distribution ($V_z \approx V_{ss} \approx 8.7$ L) and a half-life of approximately 6.5 h substantially shorter than tamoxifen. Mean absolute oral bioavailability was 69% with moderate between-subject variability (CV 21%), and negligible accumulation is expected with once-daily dosing (accumulation ratio 1.08). Body weight and CYP2D6 subtype were identified as the most relevant covariates, and a one-compartment model with first-order elimination and absorption adequately described the data. The next steps are a population PK analysis to quantify covariate effects on clearance and bioavailability, measurement of the active metabolite(s) to confirm CYP2D6 activation and establish an exposure–response relationship, and definition of a pharmacodynamic target before proceeding to Phase II. All analyses are fully reproducible from the companion interactive dashboard report, which documents every data-handling step and output in accordance with FAIR principles.

References

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